

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTEEC403 - ELECTRICAL MACHINE-II

Time : 2 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is a distinguishing feature of a synchronous machine?
a) Slip rings are not required b) It runs at constant speed irrespective of load
c) It requires external excitation at start d) It has a high starting torque
- The frequency of rotating MMF wave in a 4-pole machine running at 1500 rpm is:
a) 25 Hz b) 50 Hz
c) 100 Hz d) 200 Hz
- 3 Which of the following is the main cause of third harmonic in generated EMF?
a) Mechanical imbalance b) Slot harmonics
c) Non-sinusoidal flux distribution d) Armature reaction
- 4 The pitch factor becomes unity when coil span is:
a) 60° b) 90°
c) 120° d) 180°
Armature reaction at unity power factor in a synchronous generator is:
a) Magnetizing b) Demagnetizing
c) Cross-magnetizing d) Zero
- 6 In two-axis theory, the machine parameters are divided along:
a) Armature and rotor axes b) Direct and quadrature axes
c) Magnetic and electric axes d) Stator and rotor winding axes
- 7 Slip of a 4-pole 50 Hz induction motor running at 1440 rpm is:
a) 2% b) 4%
c) 6% d) 10%

- 8 Cogging in induction motor is due to:
 a) Low supply voltage b) Magnetic locking between stator and rotor teeth
 c) High rotor resistance d) Misaligned shaft
- 9 Single-phase induction motors are not self-starting because:
 a) Starting torque is zero b) Slip is too high
 c) High friction loss d) Rotor is unbalanced
- 10 Which method is typically used to determine equivalent circuit parameters of a single-phase induction motor?
 a) Hopkinson's test b) Open circuit and blocked rotor tests
 c) Swinburne's test d) Load test only
- Permanent Magnet AC motors differ from induction motors in that:
 a) They don't need stator winding b) They don't require magnetizing current
 c) They use slip rings d) They require DC excitation
- 12 Which of the following is true for an AC servomotor?
 a) Has low rotor inertia and high torque b) Uses a permanent magnet rotor
 c) Operates at constant speed d) Is self-synchronous

Q2. Solve the following.

Explain the constructional differences between a synchronous machine and an induction machine with neat diagrams. 6

Explain how a rotating magnetic field is produced in a three-phase AC machine. Illustrate with phasor diagrams. 6

Q3. Solve the following.

A) What are the causes of harmonics in generated EMF? Suggest methods to minimize them. 6

B) A 3-phase, 20-pole, 50Hz alternator has single-layer winding with full-pitch coils. The coils are connected in 60° phase group and each coil is having six turns. If the flux per pole is 0.025 Wb, determine the rms value of emf induced per phase. 6

Q4. Solve Any Two of the following.

A) Explain the concept of armature reaction in synchronous machines at different power factors with neat phasor diagrams. 6

B) What is meant by parallel operation of synchronous generators? Explain the conditions necessary for synchronizing a generator to an infinite bus. Describe the synchronization process using lamps or a synchroscope. 6

C) Determine the synchronising power and synchronising torque per mechanical degree of rotor displacement at no-load of a 2.5 MVA, three-phase, six-pole synchronous generator connected to 6000V, 50Hz bus-bars. The synchronous reactance of the machine is 5 ohm per phase. 6

Q5. Solve Any Two of the following.

A) Derive the expression for torque developed in a three-phase induction motor as a function of slip. Explain how torque varies with slip and identify the condition for maximum torque. 6

- B) Illustrate cogging and crawling in induction motors. How can these effects be minimized? 6
- C) A 500 HP, 3-Phase, 440 V, 50 Hz induction motor has a speed of 950 rpm at full load. The machine has 6 poles. Calculate the slip. How many complete alternations will the rotor emf make per minute. 6

Q6. Solve Any Two of the following.

- A) Why is a single-phase induction motor not self-starting? How is this issue overcome in practice? 6
- B) What is an AC servomotor? Explain its key features and applications in control systems. 6
- C) Discuss the construction and working of a linear induction motor. Mention two major industrial applications. 6

*** End ***